Lecture notes: Context-Free Grammars

A context-free grammar is G = (V, T, P, S): variables V, terminals T, productions P of the form A → α with A ∈ V and α ∈ (V ∪ T)*, and a start symbol S.
Example: S → aSb | ε generates { a^n b^n : n ≥ 0 }, a language that no finite automaton accepts.

Derivations and parse trees

A leftmost derivation always rewrites the leftmost variable. Each derivation corresponds to a parse tree whose yield is the derived string. A grammar is ambiguous if some string has two different parse trees, e.g. E → E + E | E * E | id for the string id + id * id.

Normal forms

Chomsky Normal Form allows only A → BC and A → a. Converting to CNF: remove ε-productions, unit productions and useless symbols, then break long right-hand sides. Greibach Normal Form requires every production to start with a terminal: A → aα.